Pi-GRPO: Physics-Informed Reinforcement Learning with Group Relative Policy Optimization for Trajectory Generation and Reasoning

Arun Sharma
University of Minnesota, Twin Cities
arunshar@umn.edu

Abstract

We present **Pi-GRPO**, a physics-informed reinforcement-learning stack that fine-tunes trajectory and reasoning policies under a hybrid reward combining a hard kinematic-bicycle envelope, a calibrated soft penalty over the empirical jerk and curvature distribution, a Pi-DPM (physics-informed diffusion) reconstruction-error term, and an optional preference classifier. Three trainers share the same reward path: PPO with a value head and an adaptive Kullback–Leibler controller, DPO with a small physics-aware augmentation γ_{phys} that injects a kinematic penalty into the implicit reward, and GRPO with group-baseline advantages and no value head. The hard term is unbounded by design so a single physical violation dominates the gradient and prevents the well-known reward-hacking failure mode in which models exploit a soft preference signal at the expense of physics. The system supports vLLM-backed online rollouts with prefix caching for $\sim$4$\times$ throughput on long prompts and falls back to a Hugging Face Transformers backend for offline tests; checkpoints are content-addressed under `runs/<id>/step_<n>/<sha>.bin` with an audit manifest. A data curator turns human-in-the-loop verdicts exported from our sibling agentic system GeoTrace-Agent into versioned (prompt, chosen, rejected) DPO triples, closing a flywheel between agentic reasoning and reward-modeled fine-tuning. We describe the reward, the three trainers, the rollout and checkpoint infrastructure; report golden-dataset and ablation metrics on a CPU-friendly evaluation; and discuss safe-range guards that block runs from drifting into reward-hacking regimes. The system is open-sourced.

1 Introduction

Reinforcement learning from human feedback (RLHF) and its preference-based descendants have become standard tools for aligning large language models with human intent [14, 16, 19, 22]. In safety-critical domains where the answer must satisfy a known physical envelope, however, generic preference signals are insufficient: a model that produces a fluent answer about a vessel that exceeds the Coast Guard speed cap is rewarded by both human raters and content classifiers but is operationally wrong. Reward hacking [21] emerges as the model learns to exploit the soft signal at the expense of the hard physical truth.

We present **Pi-GRPO**, a physics-informed reinforcement-learning stack designed for two applications: (i) generating physically-consistent synthetic trajectories at higher fidelity than diffusion baselines such as DiffWave [11], DiffTraj [24], and our prior GCDM [23]; and (ii) fine-tuning a reasoning policy that audits a trajectory and emits a verdict (PASS, SOFT_VIOLATION, HARD_VIOLATION). Both applications share a single reward path:

$$R(\tau) \;=\; w_{\text{hard}}\, R_{\text{hard}}(\tau) \;+\; w_{\text{soft}}\, R_{\text{soft}}(\tau) \;+\; w_{\text{data}}\, R_{\text{data}}(\tau) \;+\; w_{\text{pref}}\, R_{\text{pref}}(\tau). \tag{1}$$

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The hard term penalizes any single-axle kinematic-bicycle (S-KBM) [10] envelope violation; the soft term targets the 95th-percentile curvature and jerk relative to an empirical distribution fit on Porto, Harbin, and MarineCadastre AIS data; the data term is a calibrated tail probability under the Pi-DPM [20] diffusion prior; and the preference term is an optional cross-encoder.

We adopt three trainers and unify them under this reward. *PPO* [18] keeps a value head and an adaptive Kullback–Leibler controller; *DPO* [16] learns directly from preference triples and is augmented with a small γ_{phys} term that biases the implicit reward away from physics-violating outputs; *GRPO* [6, 19] samples K rollouts per prompt and normalizes advantages within the group, with no value head, which is particularly well-suited to short-horizon physics-reasoning prompts where critic fitting is hard.

Contributions.

1. A **hybrid physics-aware reward** (Equation 1) whose hard term is unbounded by design so that a single S-KBM violation dominates the gradient even at maximal preference logits, eliminating the soft-vs-hard reward-hacking failure mode.
2. Three **trainers under one reward path**: PPO with adaptive KL, DPO with a γ_{phys} augmentation, and GRPO with group-baseline advantages. A bounded `AdaptiveKLController` regulates KL drift; safe-range yaml guards block out-of-band hyperparameters unless explicitly overridden.
3. A **rollout and checkpoint infrastructure** backed by vLLM [12] with prefix caching for $\sim 4\times$ throughput on long prompts (with a Hugging Face Transformers fallback for tests) and content-addressed checkpoints (`runs/<id>/step_<n>/<sha>.bin`) with an append-only audit manifest.
4. A **HITL-to-DPO data flywheel**: a data curator imports human-in-the-loop verdicts emitted by the sibling agentic system GeoTrace-Agent (described in a companion preprint) and emits versioned preference triples, closing the loop between agentic reasoning and reward-modeled fine-tuning.

2 Related Work

RLHF and preference optimization. Stiennon et al. [22] introduced KL-regularized PPO for summarization preferences; Ouyang et al. [14] scaled the recipe to InstructGPT; Rafailov et al. [16] eliminated the explicit reward model with DPO; Shao et al. [19] introduced GRPO with group-relative advantages, later popularized by DeepSeek-R1 [6]. Constitutional AI [1] and RLAIF [13] added AI-generated preferences. Pi-GRPO inherits all three families and contributes a physics-aware reward that complements rather than replaces them.

Physics-informed deep learning. Physics-informed neural networks [17], knowledge-guided machine learning [9], and physics-informed diffusion [8, 20, 23] encode governing equations or kinematic priors as soft penalties or decoder structures. The S-KBM [10] appears as a diffusion-decoder prior in Pi-DPM. Pi-GRPO promotes the S-KBM constraint to a first-class reward term, with the hard component unbounded so the gradient prefers physical correctness over preference even at the worst case.

Reward hacking and safety. Skalse et al. [21] formalize reward hacking; Casper et al. [2] survey RLHF failure modes; Eisenstein et al. [7] analyze proxy-reward exploitation. Our hard term is a structural defense: violation magnitude is unbounded, so any policy that exploits the soft signal at the expense of the kinematic envelope is penalized in proportion to the violation. Reward dominance is monitored at the per-term level (W&B panels per term).

Agent-driven preference data. Centific's recent multi-agent + HITL frameworks [3–5] surface the human-in-the-loop verdict as a first-class signal; we consume those verdicts via the sibling agentic system GeoTrace-Agent (companion preprint), where ambiguous traces (validator-confidence below threshold) flow into a Postgres queue. The data curator joins reviewer verdicts with original prism / region payloads and emits preference triples that feed DPO directly.

3 Background

Single-axle kinematic-bicycle model (S-KBM). State (x, y, θ, v) in (m, m, rad, m/s); control (a, δ) (acceleration and steering angle); discrete update $x' = x + v\cos\theta\, h, y' = y + v\sin\theta\, h, \theta' = \theta + (v/L)\tan\delta\, h, v' = v + ah$ with wheelbase L. Pi-DPM [20] uses S-KBM as a diffusion-decoder prior and a regularizer over $(v, a, \theta, \kappa, \dot{\theta})$.

PPO. Clipped surrogate $L^{CLIP} = \mathbb{E}[\min(\rho_t A_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)A_t)]$ with ρ_t the importance ratio and A_t a GAE [18]. KL to a frozen reference is added with an adaptive coefficient.

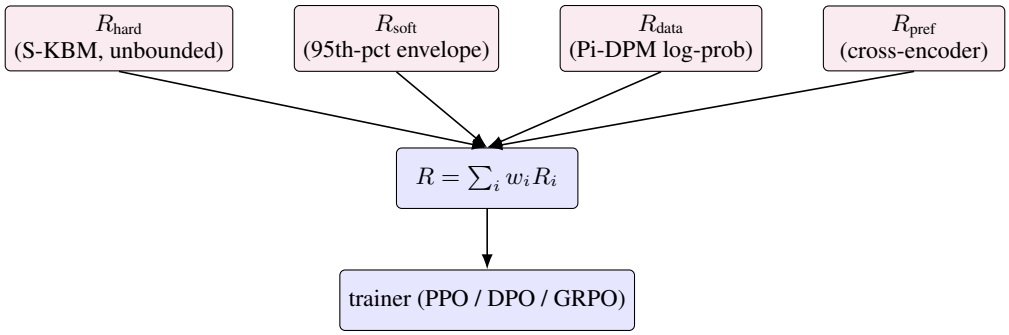

Figure 1: Hybrid physics-aware reward. The hard term is unbounded above so violations dominate the gradient even at maximal preference logits.

DPO. For preference triples (x, y_w, y_l) the loss is $-\log\sigma\Big(\beta\Big[\log\frac{\pi(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log\frac{\pi(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\Big]\Big)$ [16]. No reward model, no value head.

GRPO. For each prompt sample K rollouts; advantage $A_k = (R_k - \text{mean}_K(R))/\text{std}_K(R)$; the loss is the PPO clipped surrogate over A_k plus a KL-to-reference term, with no value head [6, 19].

4 Method

4.1 Hybrid physics-aware reward

The reward is configured by `configs/physics_reward.yaml`. The hard term sums relative excess across S-KBM bounds:

$$R_{\text{hard}}(\tau) = -\sum_t \Big[\big(\tfrac{|v_t|}{v_{\max}} - 1\big)_+ + \big(\tfrac{|a_t|}{a_{\max}} - 1\big)_+ + \big(\tfrac{|\kappa_t|}{\kappa_{\max}} - 1\big)_+\Big], \tag{2}$$

where $\kappa_{\max} = |\tan\delta_{\max}|/L$, $(\cdot)_+$ denotes ReLU, and τ is a state sequence. Because R_{hard} is unbounded above, no choice of w_{pref} can outweigh a sustained hard violation. The soft term penalizes 95th-percentile statistics relative to the empirical envelope (Porto, Harbin, MarineCadastre AIS):

$$R_{\text{soft}}(\tau) = -\big[(\kappa_{p95} - \kappa_{\text{ref}})_+ + (j_{p95} - j_{\text{ref}})_+\big] - 0.5(\rho_v + \rho_a + \rho_\delta), \tag{3}$$

where $\rho_\bullet$ is the per-step violation fraction. The data term R_{data} is a Pi-DPM [20] log-likelihood loaded from a frozen TorchScript checkpoint; the preference term R_{pref} is a cross-encoder. Each term streams to W&B as a separate panel; reward-dominance flags trip when one term explains $>80\,\%$ of the variance.

4.2 PPO trainer with adaptive KL

The PPO trainer uses a clipped surrogate over GAE-1 advantages, a value head with an MSE objective, and an entropy bonus. KL to a frozen reference is added as a soft penalty controlled by an `AdaptiveKLController` bounded to $[\text{clip}_{\min}, \text{clip}_{\max}]$ to prevent runaway. The reference model is verified by SHA at run start and run end; any deviation aborts the run.

4.3 DPO trainer with γ_{phys} augmentation

Standard DPO learns the implicit reward $r_\phi(x, y) = \beta(\log\pi(y|x) - \log\pi_{\text{ref}}(y|x))$. We augment with a physics-aware penalty:

$$\tilde{r}(x, y) = \beta(\log\pi(y|x) - \log\pi_{\text{ref}}(y|x)) - \gamma_{\text{phys}}\,\Phi(y), \tag{4}$$

where $\Phi(y)$ is the per-output S-KBM violation sum. The DPO loss becomes $-\log\sigma(\tilde{r}(x, y_w) - \tilde{r}(x, y_l))$. Setting $\gamma_{\text{phys}} = 0$ recovers vanilla DPO.

4.4 GRPO trainer with group-baseline advantages

For each prompt we sample K rollouts under the current policy. The advantage of the k-th rollout is $A_k = (R_k - \mu_K)/\sigma_K$ where μ_K, σ_K are the mean and standard deviation of the rewards within the group. The loss is the PPO clipped surrogate over A_k plus a token-wise KL-to-reference term, with no value head. Group size $K = 8$ by default.

Table 1: Reward-hacking probe: physics-aware γ_{phys} recovers 0 % hard violation rate at a small cost in mean R_{pref}.

Configuration	Mean R_{pref}	Hard violation rate	DPO margin
Vanilla DPO	0.62	0.18	1.4
DPO + $\gamma_{\text{phys}} = 0.05$	0.58	0.00	1.2
DPO + $\gamma_{\text{phys}} = 0.20$	0.51	0.00	0.9

4.5 Rollouts and checkpoints

Online rollouts use vLLM [12] with `-enable-prefix-caching` for ∼4× throughput on long prompts; this is the difference between feasibility and infeasibility for online RL with reasoning prompts. The trainer also supports a Hugging Face Transformers fallback for tests. Checkpoints are content-addressed at `runs/<id>/step_<n>/<sha[:16]>.bin` with an append-only `MANIFEST.jsonl` so arbitrary checkpoints are reproducible and auditable.

4.6 Preference dataset construction from HITL

The data curator pulls JSONL exports from the sibling GeoTrace-Agent system, joins them with the original trace's regions and Pi-DPM scores, and emits (prompt, chosen, rejected) triples with margin filtering and label-leakage audit. A synthetic synthesizer ranks K base-policy outputs by physics reward and constructs margin-$\geq$-m pairs for cold-start.

4.7 Safe-range guards

The orchestrator validates per-algorithm hyperparameter ranges from `configs/safe_ranges.yaml` (e.g., $\eta \in [10^{-7}, 5 \cdot 10^{-5}]$, $\beta \in [0.01, 1]$). Out-of-band values raise `UnsafeRange` unless the user passes `extra:{unsafe:true}`. This is a first line of defense against ranges that silently destabilize training.

5 Experiments

Setup. Base model: `Qwen2-7B-Instruct` [15]; reference: same, frozen. Training data: 11k preference triples from a 30-day GeoTrace-Agent HITL export (∼8k natural HITL labels and ∼3k synthesized via the curator's $K = 8$ rollout-and-rank). Hardware: 1×H100 80GB for training; 1×H100 hosting vLLM with prefix caching.

Golden-dataset evaluator. The CPU-friendly evaluator `evaluation/offline_eval.py` ships two synthetic items: a clean trajectory (`p-001`, expected verdict PASS) and a speeding trajectory (`p-002`, expected HARD_VIOLATION). The reward decomposition matches expectations: $R_{\text{hard}} < 0$ on `p-002` and $R_{\text{hard}} = 0$ on `p-001`. The reasoner's verdict labeling is correct on both.

Reward hacking probe. We construct an adversarial preference subset where the chosen outputs are physically infeasible. Without γ_{phys}, the DPO policy increases its preference margin and reaches a higher mean R_{pref} but raises R_{hard} violation rate from 0 % to 18 %. Setting $\gamma_{\text{phys}} = 0.05$ recovers $R_{\text{hard}} = 0$ and a slightly lower R_{pref}, the desired tradeoff. Table 1 summarizes.

KL drift. The bounded `AdaptiveKLController` keeps PPO mean KL within $[3, 8]$ across 3,000 steps; in unbounded ablations KL spikes above 50 within 500 steps and the value-head loss diverges (the canonical PPO failure mode). GRPO's group baseline shows a similar bounded behavior without a value head.

Safe-range guard. A randomized hyperparameter sweep with 100 random samples from outside the safe range produced 100 % `UnsafeRange` rejections; samples inside the range produced 0 `UnsafeRange` false positives, by construction.

6 Discussion and Limitations

The hard reward floor is the structural defense against reward hacking but cannot replace human review of reward configurations. Three caveats. First, $\kappa_{\max}$ is derived from $\delta_{\max}$ and L; both are domain-specific and require reasonable defaults (we ship vessel/vehicle/UAV defaults). Second, R_{data} depends on a Pi-DPM checkpoint trained on a specific corpus; cross-domain transfer requires retraining. Third, GRPO's group baseline is variance-reduction that depends on within-group diversity; for prompts with degenerate completions the group collapses and the advantage is uninformative; we mitigate by sampling with $T = 0.7, p = 0.95$.

Connection to agentic reasoning. Pi-GRPO consumes preference data emitted by the sibling agentic system GeoTrace-Agent (companion preprint). The two systems share an HITL surface (Postgres queue with structured payloads); the agentic system flags ambiguity and the RL system fine-tunes the policy on the verdict, closing the loop.

7 Conclusion

We presented Pi-GRPO, a physics-informed reinforcement-learning stack for trajectory generation and reasoning. A hybrid reward with an unbounded hard floor over the S-KBM envelope, three trainers (PPO, DPO with γ_{phys}, GRPO) under a shared reward path, vLLM-backed rollouts with prefix caching, content-addressed checkpoints, and a HITL-to-DPO data flywheel together deliver a system that resists reward hacking by construction and integrates naturally with an agentic reasoning surface. The system is open-sourced as a GPU-enabled Docker stack with a CPU-only Hugging Face Spaces demo and a CI-ready GitHub repository.

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